Cultivating Computational Equity: Building a Collaborative Research Computing Community in California's Central Valley

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Abstract

Institutions in California's Central Valley face persistent barriers to research computing, including limited local expertise, low utilization of national resources, and a lack of coordinated training. CENVAL-ARC, a regional collaboration among UC Merced and nearby CSU campuses, was established to address these challenges by lowering technical barriers and fostering shared infrastructure and community. This paper presents an overview of early activities—needs assessment, skills training, and a regional symposium—and highlights lessons from building cross-institutional coordination. The approach offers a replicable model for other regions aiming to strengthen research computing support at under-resourced institutions through collaborative, place-based strategies.

CCS Concepts

• **General and reference** → **Surveys and overviews**; *General literature*; • **Human-centered computing** → *Accessibility theory, concepts and paradigms.*

Keywords

High Performance Computing (HPC), Computational access, Underserved regions, Hispanic-Serving Institutions (HSIs), Cross-institutional collaboration, Equity in computing, Inclusive research ecosystems, Community building, Regional partnerships, Sustainable infrastructure, CENVAL-ARC, Central Valley

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1 Introduction

The evolution of Research Computing and Data (RCD) has transformed scientific inquiry, enabling researchers to address complex questions at unprecedented scales[5]. Despite these advancements, equitable access to computational resources remains concentrated within well-resourced institutions. As highlighted by National Science Foundation (NSF)'s Missing Millions report[1]: democratizing computing access to bridge the digital divide, the Central Valley Accessible Research and Computational Hub (CENVAL-ARC) is driven by the recognition that researchers and students without extensive computing skills are often excluded from utilizing large-scale computing investments. Even as cloud technologies transform the academic enterprise landscape, barriers to cloud adoption are both technical and financial, especially for non-R1s and smaller institutions[3]. One participant said, "I wanted to scale beyond my local campus resources to cloud resources like Amazon, but who's paying for Amazon? No one." This paper describes CENVAL-ARC's approach to building regional partnerships by identifying common computational challenges and opportunities.

2 Assessing Computational Resource Gaps in the Central Valley

A foundational step in establishing CENVAL-ARC was identifying shared challenges among Central Valley institutions: University of California Merced (UCM) and California State University (CSU) campuses in Fresno, Sacramento, and Stanislaus. In 2024, we observed scant resource use of NSF's Advanced Cyberinfrastructure Coordination Ecosystem: Services and Support (ACCESS) at these institutions, with a snapshot of allocations as a percentage of total ACCESS CPU allocations as follows 1:

- CSU Fresno: A 0.04% allocation, 3 PI projects, and 6 users
- CSU Sacramento: A 0.03% allocation, 2 projects, and 4 users

- UCM: A 0.02% allocation, with 13 faculty members accessing resources
- CSU Stanislaus: No allocation, no project, and no users

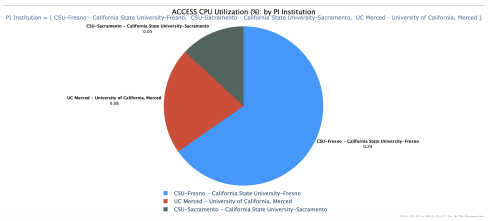


Figure 1: ACCESS CPU utilization percentage (%) by PI institution in 2024

Despite being eligible and potentially in need of High Performance Computing resources, researchers at these four institutions are significantly underutilizing national supercomputing resources. This gap highlights the importance of intermediate clusters that offer both computational resources and researcher support. Since small- and mid-scale HPC clusters share many features with large-scale HPC (e.g. schedulers, software, job types) such as those underpinning ACCESS, building skills locally is a key step in allowing researchers to access resources at multiple scales. CENVAL-ARC, designed with accessibility and training in mind, aims to help address this gap.

2.1 Challenges

The common challenges across institutions quickly emerged in detail:

- **Limited Local Expertise**: Many campuses reported insufficient specialized support personnel, such as research computing facilitators and HPC system administrators. This shortage often resulted in underutilization of available computing resources and hindered the scaling of research projects.
- **Insufficient Computational Training**: Insufficient Computational Training: Students and faculty across institutions faced similar gaps in training opportunities and resources, limiting their ability to fully leverage advanced computational tools and techniques, such as user interaction via the command line interface, parallel computing, machine learning, and data-intensive methodologies.
- **Access Barriers to National Resources**: As highlighted in the white paper "The Role of Regional Organizations in Improving Access to the National Computational Infrastructure"[6], national HPC centers often excel at community engagement but face limitations in reaching every campus, conference, or workshop. In contrast, regional organizations benefit from geographical proximity, local trust, and easier access to colleges and universities, and local user communities.

2.2 Opportunities

Simultaneously, multiple opportunities emerged that made regional collaboration attractive and feasible:

- **Connected communities**: The demographic similarity of student populations across these predominantly Hispanic Serving, underrepresented minority (URM) institutions, including first-generation college students and students eligible for Pell Grants[7], positions them uniquely for collaborative initiatives to address diverse educational and research needs. Notably, UCM is the only doctorate-granting institution in California's Central Valley.

Table 1: Demographic Overview of CENVAL-ARC Partner Institutions (Fall 2022)

Institution	Total UnderGrad	% Under Represented Minorities	% Hispanic	% First-Gen	% Pell Eligible
UC Merced	8,300+	60%	55%	66%	58%
CSU Sacramento	28,400	44%	36%	31%	43%
CSU Stanislaus	8,000	64%	61.4%	71.5%	58.2%
CSU Fresno	21,000+	76%	58%	N/A	62%

- **Geographic Proximity**: The geographic proximity of these institutions in California's Central Valley facilitates collaboration and STEM research advancement. This more easily facilitates in-person workshops, joint training sessions, and sharing of local computational expertise. All locations are within a two-hour drive of UC Merced (Fig. 2).



Figure 2: Geographic proximity of CENVAL-ARC partner institutions in California's Central Valley

3 Laying the Groundwork: Skills Development and Regional Coordination

To address the challenges outlined above, an early CENVAL-ARC activity included hosting an online Introduction to Unix training session in February 2025, attended by over 30 participants from multiple campuses. The session was designed to help faculty and students begin working with command-line environments—an essential entry point to using HPC systems[8].

In conversations with stakeholders from participating institutions, the divide between computation for research and computation for instruction was highlighted. One symposium participant noted, "democratization of access is a core role of education in computing. In order to do research you need access to these clusters, trainings and resources. Through this conference, I have learned a lot about accessible resources." Integrating computational approaches into the classroom builds a pathway for students to engage in computationally-focused research, both as undergraduates and as graduate students.

4 CENVAL-ARC Symposium: Laying the Groundwork for Regional Cooperation and Training

The CENVAL-ARC Symposium at UC Merced was a multi-day, multi-mode event featuring sessions on HPC and high-throughput computing (HTC), approaches to research computing and data-intensive workflows, as well as computational research and training. The event brought together faculty, graduate and undergraduate student researchers who require computational power beyond their laptops or local resources[2].

4.1 Learning Objectives

The program's learning objectives for attendees were to:

- Understand the regional and national landscape of RCD.
- Be familiar with the Campus Research Computing Consortium (CaRCC) capabilities model in order to assist Central Valley institutions in evaluating and enhancing their RCD support systems and capabilities[4].
- Develop basic HPC skills, including:
  - Build foundational skills in command-line, environments
  - Edit and implement sample Bash scripts,
  - Log into and utilize Open Science Grid (OSG) systems.

Virtual sessions further reinforced practical skills and offered additional hands-on opportunities and deeper engagement with computational practices and tools:

- Office hours for OSG and CaRCC Cyberinfrastructure planning allowed direct, personalized support for attendees addressed specific questions arising from the hands-on workshops.
- Pre-symposium training for "Introduction to Unix" familiarized participants with the Unix command, and was attended by over 30 participants from multiple campus.
- UC Merced computational faculty led discipline-specific sessions that helped participants apply their new skills to their own research areas.
- The session on best practices in writing research software complements practical computational skills by emphasizing reproducibility, maintainability, and broader research impact.

4.2 Outreach and Engagement Strategy

With no prior regional event of this kind, building attendance for the inaugural symposium required a grassroots strategy. Organizers used personalized email campaigns and reciprocal outreach through institutional leaders and faculty contacts to grow awareness. Established personal and professional networks were also key, especially for the organizers.

4.3 Interdisciplinary Expertise and Research Themes at the Symposium

The focus areas represented at the symposium spanned a wide range of disciplines, with significant emphasis on computational science, AI/ML (artificial intelligence/machine learning), HPC, and data science. There was a strong interdisciplinary theme.

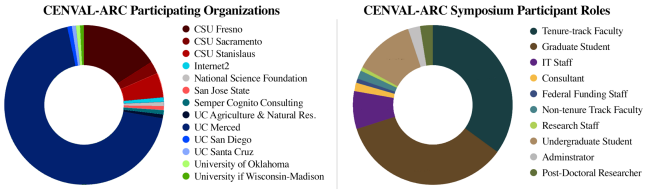


Figure 3: Institutions and Institutional Roles of CENVAL-ARC Symposium Participants

4.4 Participant Feedback

Participant feedback highlighted the symposium's impact in raising awareness of computational resources and fostering regional collaboration. Attendees reported new insights, stronger connections, and a clear interest in continued engagement—early signs of momentum toward addressing shared computational challenges. As one panelist noted, "democratization of access is a core role of education in computing. In order to do research you need access to these clusters, trainings and resources. Through this conference, I have learned a lot about accessible resources."

An attendee shared via email, "Thank you very much one more time for holding such a unique conference on your campus. If there is one next year, I will surely attend." Another attendee succinctly commented, "It was a good symposium. Thank you!" These responses signal growing collaboration and confirm the symposium's role in addressing regional computational needs.

5 Reflections and Future Work

To build upon the early momentum generated by the CENVAL-ARC initiative, ongoing efforts will prioritize sustained community engagement, inclusive access, and continuous improvement. Key strategies include:

- **Regular Community Engagement**: CENVAL-ARC will maintain quarterly meetings to foster ongoing collaboration, knowledge exchange, and continuous feedback to help sustain the dialogue initiated at the symposium, providing participants with consistent opportunities to connect, learn, and share resources.
- **Expanding Computational Training** : Future training programs will be responsive to the evolving needs of researchers, emphasizing hands-on experience in cloud computing, data management, and software development practices.
- **Enhanced Inclusivity and Accessibility**: To ensure that CENVAL-ARC's resources are equitably accessible, the initiative will actively engage underrepresented researchers and students through targeted outreach and tailored programming.
- **Capturing Richer Demographic Data** Figure 3 shows participating institutions and roles. To evaluate online versus in-person training, we need richer participant data—including demographics across formats, learner assessments, learning perception, and higher-order thinking measures.

6 Conclusion

Moving forward, CENVAL-ARC will strengthen collaborations and address evolving computational needs through responsive programming. Gathering of data more systematically will allow CENVAL-ARC investigators to better serve CENVAL-ARC community, while increasing the value of annual reports and academic papers, such as this. Stakeholder engagement will inform future directions, ensuring that initiatives remain relevant and beneficial to the evolving landscape of computational research.

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